BIOCHEMISTRY CUMULATIVE EXAM

October 3, 1992

Site selective mutagenesis provides a means whereby one amino acid can be selectively replaced by another amino acid at a single site in a protein through the manipulation of genetic material.

a. (10 points) In general terms, explain what must be accomplished before site selective mutagenesis can be useful in study of protein structure and function.

b. (20 points) Some proteins are difficult to express in E. coli as they are toxic to the host cell, sometimes for unknown reasons. Describe in general terms at least two possible ways to overcome this problem such that "problem proteins" might be expressible in E. coli.

c. (30 points) Assuming that all criteria for site selective mutagenesis can be met, an appropriate choice for the amino acid to be changed must be made. For each of the situations shown below, suggest an appropriate change to result in the desired (described) effect.

example: replacement of a large hydrophobic residue with one of similar size — Leu → Ile

c1. replacement of a nonpolar aromatic residue with a polar aromatic residue

c2. replacement of a residue capable of forming disulfide bonds with a residue of comparable size but unable to form disulfide bonds

c3. replacement of a positively charged residue with another of like charge

c4. replacement of a negatively charged residue with an uncharged residue of similar size

c5. replacement of an hydroxyl containing amino acid with one of comparable size to determine whether size or H-bonding capability is important at that site

d. (40 points) A protein, P, binds a small molecule, M, according to the equation below.

$$P + M \Longleftrightarrow PM$$

Several single site mutants of P have been generated and the K_{eq} for the binding of M to P determined in each case. How would you proceed to determine whether multiple changes made in the protein through site selective mutagenesis act independently or synergistically.